

1. Consider the steady state heat transfer through a rectangular plate of size as mentioned in the figure. The top and the bottom walls are adiabatic. The left wall is at 1000 K, and the right wall is at 500 K. Find the steady-state temperature distribution inside the plate and plot the temperature contour using a uniform grid of cell size 0.01 m. Next, plot the temperature distribution at $y = 0.5$ m along x by extracting the data from the simulation. To find the temperature contour, solve the equations using the following methods and compare the number of iterations required in each method for an accuracy of 0.1 K
- Jacobi Method
 - Point Gauss-Seidel Method

